

Third Degree Price Discrimination in the Hotelling's Linear City



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Introduction

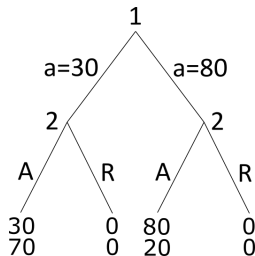
- Price discrimination and monopoly.
- Price discrimination and oligopoly:
 1. Global competition models (Cournot, Bertrand-Edgeworth);
 2. **Local-Spatial competition models (Line, Circle, Spokes, Networks).**
- What happens to firms' profits, consumers' surplus and social welfare?

About Sequential Games

- Simultaneous games and the Nash-Equilibrium definition are assumed common-knowledge, if not see (Mas-Colell et al. 1995).
- Unlike a simultaneous game, a sequential game is a game where a player selects its action before the other, and the subsequent player is informed of that choice before making its own decisions. They are typically represented by decision trees, mapping out all possible sequences of play.
- In sequential games with perfect information a *subgame perfect Nash-equilibrium* (SPNE), which is a set of strategies inducing a NE in each subgame, can be determined through *backward induction*. Such process is aimed at working from the end of the game back to the start to identify optimal strategies in each subgame.
- Examples: Tic-Tac-Toe, Checkers or Chess.

A Light Example: The Bargaining Game

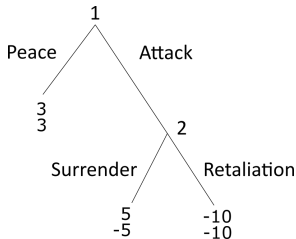
Suppose that two agents have 100 euros to split, and they bargain according to the following procedure. First, agent 1 proposes an allocation of the money: $(a, 100 - a)$ where a is either 30 or 80. Then agent 2 accepts the proposal of agent 1 or rejects it. If 2 accepts it, each agent receives what 1 proposed: $(a, 100 - a)$. If 2 refuses it, the money is wasted, and each one of them receives zero: $(0, 0)$.



The couple of strategies $(a = 80, A)$ is a SPNE.

A Not So Light Example: The Nuclear War Game

Suppose now that two nuclear powers are in some form of conflicts, and they consider the use of nuclear weapons as follows. First, power 1 considers either to stay in peace or to attack. Then power 2 decides to either surrender or to retaliate. If 1 opts to stay peaceful they both get 3, while if he decides to attack, either the second power surrender yielding $(5, -5)$ or retaliate for $(-10, -10)$.



The SPNE in this case is $(\text{Attack}, \text{Surrender})$.

The Hotelling Model as a Game

- The Hotelling model involves a linear city of length 1 with evenly spaced identical consumers traveling and consuming from two identical firms.
- The two firms play a sort of two-stage sequential game made of two simultaneous games, choosing first their locations and then their prices.
- The game is solved by backward induction, with firms solving for the equilibrium prices conditioning on the locations chosen, to eventually pick the actual locations of equilibrium, finally deriving the game's SPNE.
- Consumers face a transportation cost, which might be linear or **quadratic** when moving from their location to one of the firms.

where $v > 0$ is the reservation utility, $p_i > 0$ is the price charged by i and $t > 0$ is the transportation (or differentiation) cost.

One Key Assumption

- In order to simplify the framework - specifically the demand function - it is almost always assumed a v large enough for all consumers to experience a positive utility and consume one unit of good each.
- In other words, the evaluation of the good made by each consumer is so high that despite the distance, even the furthest consumer will opt to buy yielding a "Fully-Covered Market".
- It is worth noting that to reach full-market coverage one might also assume the utility of not-consuming to be minus infinity.

The Timeline of the Game and Its Equilibrium

- The two subgames are:
 1. Firms choose locations;
 2. Firms set prices.
- As noted above, the game might be solved by backward induction with firms first solving for the equilibrium prices conditioning on the locations and then solving for the equilibrium locations.
- Such process, it will be shown, yields a unique SPNE, or the unique couple of prices and locations

$$((l_i^*, l_j^*), (p_i^*, p_j^*))$$

inducing a Nash Equilibrium in each subgame.

Backward Induction: Step 2 - I

- To find the equilibrium prices conditional on locations we proceed as follows:

- The indifferent consumer $\hat{x}$ between firm i and j is found equating $u(x)$ when buying from i and $u(x)$ when buying from j

$$v - p_i - t|x - l_i|^2 = v - p_j - t|x - l_j|^2$$

and solving for x

$$\hat{x} = \frac{l_i + l_j}{2} - \frac{p_i - p_j}{2t(l_j - l_i)} \quad (1)$$

- Since we assumed full-market coverage, or that any consumer buys a unit of good, the demand for firm i is

$$D_i = \hat{x}$$

while the one of j

$$D_j = 1 - \hat{x}$$

Backward Induction: Step 2 - II

3. Both firms, facing the usual marginal cost $c > 0$, wishes to solve the following profit maximization problems

$$\max_{p_i} \pi_i = (p_i - c) \hat{x}$$

and

$$\max_{p_j} \pi_j = (p_j - c) (1 - \hat{x})$$

which, it can be shown, yield two unique solutions recoverable through the usual FOCs.

4. Solving the system of FOCs, the equilibrium prices conditional on the locations are

$$\bar{p}_i = c + t \frac{(l_j - l_i)(2 + l_i + l_j)}{3} \quad (2)$$

$$\bar{p}_j = c + t \frac{(l_j - l_i)(4 - l_i - l_j)}{3} \quad (3)$$

Backward Induction: Step 2 - III

5. Substituting (2) and (3) into (1) we get

$$\bar{x} = \frac{2 + l_i + l_j}{6}$$

hence

$$\bar{D}_i = \frac{2 + l_i + l_j}{6}$$

$$\bar{D}_j = 1 - \frac{2 + l_i + l_j}{6}$$

Backward Induction: Step 1 - I

- Once we found the equilibrium prices conditional on the locations we seek the equilibrium locations as follows:
 - Once again firms wishes to solve the following maximization problems

$$\max_{l_i} \bar{\pi}_i = t \frac{(l_j - l_i)(2 + l_i + l_j)^2}{18}$$

$$\max_{l_j} \bar{\pi}_j = t \frac{(l_j - l_i)(4 - l_i - l_j)^2}{18}$$

which, it can be shown, yield two unique solutions recoverable through each so-called Karush-Kuhn-Tucker (KKT) associated system.

Backward Induction: Step 1 - II

2. Such KKT associated systems yields the two best-reply functions $l_i = 0$ and $l_j = 1$ and thus the system

$$\begin{cases} l_i = 0 \\ l_j = 1 \end{cases}$$

yields trivially the equilibrium locations

$$l_i^* = 0 \quad (4)$$

$$l_j^* = 1 \quad (5)$$

Without going through the KKT systems, intuitively it is enough to note that

$$\forall l_i \in [0, l_j), \quad \frac{\partial \bar{\pi}_i}{\partial l_i} < 0$$

and

$$\forall l_j \in (l_i, 1], \quad \frac{\partial \bar{\pi}_j}{\partial l_j} > 0$$

The Subgame Perfect Nash Equilibrium

- Finally by plugging (4) and (5) into (2) and (3) we are able to find the actual equilibrium prices (no more conditional on locations)

$$p_i^* = c + t$$

$$p_j^* = c + t$$

- Therefore the unique SPNE of the Hotelling game with quadratic transportation cost is

$$((0, 1), (c + t, c + t))$$

meaning maximum differentiation and increasing competition as t goes to 0 (indeed when t goes to 0, π_i^* and π_j^* goes to 0).

Third Degree Price Discrimination in a Linear City: Premise

- Price discrimination in oligopoly theory adds to the literature of monopolist's price discrimination a large collection of real-world situations worth to be analyzed with the tools of industrial organization.
- Namely, in what follows, we will deal with third degree price discrimination in a linear city model à la Hotelling where two firms are located in 0 and in 1.
- Adhering to (Zhang et al. 2019) the game develops in two stages:
 1. Firms decide whether to acquire information and to segment the market;
 2. Firms set prices.

which solved by backward induction yields, it will be shown, one or two SPNEs.

Fixed Firms' Locations and Linear Transportation Costs

- Let a unit mass of consumers interested in purchasing either 0 or 1 unit of a given good be uniformly distributed on a line of length one. Let also two firms i and j be located on the same line in $l_i = 0$ and $l_j = 1$.



- The utility function of a generic consumer x is defined as

$$u(x) = \begin{cases} v - p_i - t(x - l_i) & \text{when buying from } i \\ v - p_j - t(l_j - x) & \text{when buying from } j \end{cases}$$

where $v > 0$ is the reservation utility, $p_i > 0$ is the price charged by i and $t > 0$ is the transportation (or differentiation) cost.

- Note that in this case the transportation costs are assumed linear for ease of computation, still the results stay unchanged with quadratic transportation costs.

The Timeline of the Game and its Equilibrium

- The two subgames are:
 1. Firms decide whether to acquire information paying a cost c ;
 2. Firms set prices.
- This game is solvable by backward induction to reach a SPNE, or the couple of decisions on information acquisition and related prices inducing a NE in each subgame

$$((s_i^*, s_j^*), (p_i^*, p_j^*))$$

with $s_i^*, s_j^* \in \{I, NI\}$ where I means the firm acquires information and NI means the firm does not acquire information.

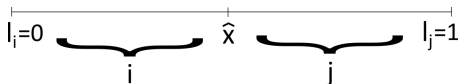
Backward Induction: Step 2 - I

- In step 2 we need to find the equilibrium prices conditional on the decision of information acquisition. Such subgame consists of 4 other subgames (sub-subgames):
 1. Both firms do not acquire information (NI,NI);
 2. One firm acquires information while the other does not (I, NI) or (NI,I);
 3. Both firms acquire information (I,I).
- In what follows we seek a NE in each of those subgames.

Backward Induction: Step 2 - II

1. (NI,NI)

When both firms do not acquire information, we deal with the usual Hotelling model with fixed locations and linear transportation costs.



We proceed then as follows:

- a. The indifferent consumer is found as usual

$$\hat{x} = \frac{1}{2} + \frac{p_j - p_i}{2t} \quad (6)$$

and assuming the market is fully covered

$$\begin{aligned}
 D_i &= \hat{x} \\
 D_j &= 1 - \hat{x}
 \end{aligned}$$

Backward Induction: Step 2 - III

- b. Both firms wishes to solve the following profit maximization problems

$$\max_{p_i} \pi_i = p_i \hat{x}$$

$$\max_{p_j} \pi_j = p_j (1 - \hat{x})$$

which it can be shown yield two unique solutions recoverable through the usual FOCs.

- c. Solving the system of FOCs

$$\begin{cases} \frac{\partial \pi_i}{\partial p_i} = 0 \\ \frac{\partial \pi_j}{\partial p_j} = 0 \end{cases}$$

Backward Induction: Step 2 - IV

the equilibrium prices conditional on the information acquisition (NI, NI) are

$$\bar{p}_i = t$$

$$\bar{p}_j = t$$

while the conditional indifferent consumer and equilibrium profits are

$$\bar{x} = \frac{1}{2} \tag{7}$$

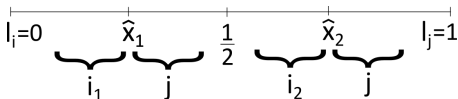
$$\bar{\pi}_i = \frac{1}{2}t \tag{8}$$

$$\bar{\pi}_j = \frac{1}{2}t \tag{9}$$

Backward Induction: Step 2 - V

2. (I,NI)

It is worth noting that when only one firm acquires information, it can be either firm i (I, NI) or firm j (NI, I). The two cases are symmetric and for simplicity we only consider the first one, (I, NI). When i acquires information the market to the left and the market to the right of $\bar{x} = \frac{1}{2}$ are segmented into two submarkets each, at $\hat{x}_1$ and $\hat{x}_2$ respectively. Firm i , having acquired information, sells in its two submarkets at two different prices p_{i1} and p_{i2} , firm j sells at a uniform price p_j



We proceed then as follows:

Backward Induction: Step 2 - VI

- a. The two indifferent consumers $\hat{x}_1$ and $\hat{x}_2$ are found imposing in (6) either p_{i1} or p_{i2} in place of p_i

$$\hat{x}_1 = \frac{1}{2} + \frac{p_j - p_{i1}}{2t}$$

$$\hat{x}_2 = \frac{1}{2} + \frac{p_j - p_{i2}}{2t}$$

then once again assuming the market is fully covered both i and j seek a solution to their profit maximization problems

$$\max_{p_{i1}, p_{i2}} \pi_i = p_{i1} \hat{x}_1 + p_{i2} \left(\hat{x}_2 - \frac{1}{2} \right) - c$$

$$\max_{p_j} \pi_j = p_j \left(\frac{1}{2} - \hat{x}_1 \right) + p_j (1 - \hat{x}_2)$$

Backward Induction: Step 2 - VII

- b. Once again, showing that a unique solution exists to both, and solving the system of associated FOCs

$$\begin{cases} \frac{\partial \pi_i}{\partial p_{i1}} = 0 \\ \frac{\partial \pi_i}{\partial p_{i2}} = 0 \\ \frac{\partial \pi_j}{\partial p_j} = 0 \end{cases}$$

the equilibrium prices conditional on the information acquisition (I,NI) are

$$\bar{p}_{i1} = \frac{3}{4}t$$

$$\bar{p}_{i2} = \frac{1}{4}t$$

$$\bar{p}_j = \frac{1}{2}t$$

Backward Induction: Step 2 - VIII

while the conditional equilibrium profits are

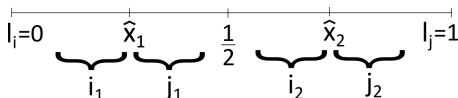
$$\bar{\pi}_i = \frac{5}{16}t - c \quad (10)$$

$$\bar{\pi}_j = \frac{1}{4}t \quad (11)$$

Backward Induction: Step 2 - IX

3. (I,I)

When both i and j acquire information the market to the left and the market to the right of $\bar{x} = \frac{1}{2}$ are still segmented into two submarkets each, at $\hat{x}_1$ and $\hat{x}_2$ respectively. But in this case both firms, having acquired information, sell in their two submarkets at two different prices p_{i1} , p_{i2} and p_{j1} , p_{j2}



We proceed as follows:

Backward Induction: Step 2 - X

- a. The two indifferent consumers $\hat{x}_1$ and $\hat{x}_2$ are found imposing once again in (6) either, p_{i1} and p_{j1} , or, p_{i2} and p_{j2} , in place of p_i and p_j

$$\hat{x}_1 = \frac{1}{2} + \frac{p_{j1} - p_{i1}}{2t}$$

$$\hat{x}_2 = \frac{1}{2} + \frac{p_{j2} - p_{i2}}{2t}$$

then once again assuming the market is fully covered both i and j seek a solution to their profit maximization problems

$$\max_{p_{i1}, p_{i2}} \pi_i = p_{i1} \hat{x}_1 + p_{i2} \left(\hat{x}_2 - \frac{1}{2} \right) - c$$

$$\max_{p_{j1}, p_{j2}} \pi_j = p_{j1} \left(\frac{1}{2} - \hat{x}_1 \right) + p_{j2} (1 - \hat{x}_2) - c$$

Backward Induction: Step 2 - XI

- b. Once again, showing that a unique solution exists to both, and solving the system of associated FOCs

$$\left\{ \begin{array}{l} \frac{\partial \pi_i}{\partial p_{i1}} = 0 \\ \frac{\partial \pi_i}{\partial p_{i2}} = 0 \\ \frac{\partial \pi_j}{\partial p_{j1}} = 0 \\ \frac{\partial \pi_j}{\partial p_{j2}} = 0 \end{array} \right.$$

Backward Induction: Step 2 - XII

the equilibrium prices conditional on the information acquisition (I,I) are

$$\bar{p}_{i1} = \frac{2}{3}t$$

$$\bar{p}_{i2} = \frac{1}{3}t$$

$$\bar{p}_{j1} = \frac{1}{3}t$$

$$\bar{p}_{j2} = \frac{2}{3}t$$

while the conditional equilibrium profits are

$$\bar{\pi}_i = \frac{5}{18}t - c \quad (12)$$

$$\bar{\pi}_j = \frac{5}{18}t - c \quad (13)$$

Backward Induction: Step 1

- In step 2 we found the equilibrium prices conditional on the information acquisition decisions, namely the equilibrium prices conditional on (NI,NI), (I,NI) and (I,I).
- Now we seek the equilibrium acquisition decisions as follows:
 1. Equations (8) to (13) give us the equilibrium profits conditional on each possible couple of information acquisition decisions.
 2. We are therefore able to construct a matrix of payoffs to easily detect Nash-Equilibria,

	NI	I
NI	$1/2t, 1/2t$	$1/4t, 5/16t - c$
I	$5/16t - c, 1/4t$	$5/18t - c, 5/18t - c$

3. Above table clearly indicates (I,I) and (NI,NI) as NEs if $0 \leq c \leq \frac{t}{36}$ while only (NI,NI) if $c > \frac{t}{36}$.

Example: Set $c = 0$ and check that (I,I) and (NI,NI) are both Nash-Equilibria.

The Subgame Perfect Nash Equilibrium I

- Above we noted that conditional to (NI, NI) the couple of equilibrium prices is (t, t) and that conditional to (I, I) is $((\frac{2}{3}t, \frac{1}{3}t), (\frac{1}{3}t, \frac{2}{3}t))$.
- Therefore if $0 \leq c \leq \frac{t}{36}$ the two SPNEs are

$$((NI, NI), (t, t))$$
$$\left((I, I), \left(\left(\frac{2}{3}t, \frac{1}{3}t \right), \left(\frac{1}{3}t, \frac{2}{3}t \right) \right) \right)$$

while if $c > \frac{t}{36}$ the unique SPNE is

$$((NI, NI), (t, t))$$

The Subgame Perfect Nash Equilibrium II

- In the case of a small enough cost of information acquisition (namely $0 \leq c \leq \frac{t}{36}$), it is not surprising that both firms price discriminating can be supported as a SPNE. When a firm's weak market is its rival's strong market, price discrimination allows the firm to steal its rival's consumers without lowering price to its own loyal consumers. As a result, although both firms' price discrimination lowers the firms' profits, each firm has an incentive to price discriminate unilaterally. Still, both firms prefer the uniform pricing SPNE to the price discrimination SPNE, suggesting that no firm acquiring information is the SPNE more likely to be reached.

The Subgame Perfect Nash Equilibrium III

- Price discrimination affects consumer welfare through two channels. First, it induces some consumers (close to the middle) to buy from a distant store. This increases consumers' transportation cost, hurting consumers. Second, it intensifies price competition between firms, meaning lower prices for consumers. This benefits consumers. Overall, it can be proved that price discrimination benefits both consumers who buy from a close firm and who buy from a distant firm, raising overall consumer surplus.
- Still, public policy intervention to favor price discrimination is a difficult call as it can be shown that price discrimination decreases profits more than the increase in consumers' welfare, yielding a lower social surplus.

A Final Remark

- Is it possible to gradually migrate from third degree price discrimination to first degree (perfect) price discrimination?
- We will show that this is not only possible but likely to happen when firms are increasingly able to collect data about consumers' preferences.
- Next lecture will deal with such phenomenon which is at very center of current EU Competition Law debates.

References

Mas-Colell, Andreu, Jerry R. Green and Michael D. Whinston. 1995. "Microeconomic Theory". Oxford: Oxford University Press.

Zhang, Tong, Yixue Huo, Xin Zhang and Jie Shuai. 2019. "Endogenous third-degree price discrimination in Hotelling model with elastic demand". Journal of Economics, 127 (2): 125-45.